The smooth 23×80 model: first results

Accuracy, calibration, within-bin and edge tests, sensitivity

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In brief

1. **It works, and it is the best model we have.** On the 1,000 test spectra the smooth model ($J = 40$ splines per function, threshold factor) is as accurate as the best alternative (Poisson deviance per bin 1.566 below and 1.100 above 555 GeV, against 1.559 and 1.105 for the free per-bin fit with Poisson weights), with an exact fit uncertainty of 2.9% of the noise. With the discrepancy its pulls have s.d. 1.004 (1.011 below, 1.001 above 555 GeV), 95.4% within ± 2, and are calibrated at every degree of cancellation ($\langle z^2 \rangle$ = 0.97–1.04).
2. **It describes the inside of a bin.** Fitted on 30-GeV bins, it predicts how each splits into two 15-GeV bins to within the noise above about 360 GeV, where the free fit cannot (split χ^2 up to 30). Below 345 GeV the threshold structure is finer than 30 GeV and the split is predicted less well.
3. **The threshold factor matters**; extrapolation above 1455 GeV does not work.
4. **A finding on the way: weighting by the observed counts biases fits low.** All earlier fits (the network fine-tune, the free fit) used weights $1/(q(t+q))$; with tens of events per bin at high mass this biases them low by a measurable amount. The smooth model uses the model's own Poisson variances (iterated), and passes a toy test with known answers (pull s.d. 0.97, mean +0.06).

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1 Implementation and the toy test

The model is in `dihiggs/smooth.py` (`SplineBasis`, `SmoothHEFT`); the plan is `docs/smooth/smooth_plan.pdf`. Fitting the 3,989 training spectra (281,044 rows including 1,814 restored empty bins) takes about a minute on a laptop CPU.

Toy test (`code/smooth_toy.py`). Three known smooth coefficient functions, 600 spectra of 40,000 equal-weight events in 15-GeV bins. (i) For fixed smoothness the fit agrees with a direct dense solve to 6×10^{-14}. (ii) With the smoothness chosen by the evidence, the fitted functions agree with the truth within their quoted uncertainties: pulls over 80 points $\times$ 3 functions have mean $+0.06$ and s.d. 0.97.

Weighting by the observed counts biases a fit low

In the first version of the toy test the pulls had mean -0.93, worst at high mass. The cause is classic: weighting each bin by $1/(q\,n)$, with n the observed count (Neyman's χ^2), gives downward fluctuations larger weights, so the fit is pulled low by about one event per bin; with $\sim$20–50 events per bin at high mass and 3,989 spectra pooled, that is many standard deviations of the fit. The cure is to weight by the model's own Poisson variance, $q\max(f, q)$, and iterate (`fit_poisson`); at convergence this is the Poisson maximum-likelihood fit. Where the model predicts a negative value the observed count is used, and each iteration moves half-way, which avoids oscillation. The network fine-tune and the free fit used the biased weights; table 1 shows the effect.

2 Accuracy

Because the Neyman χ^2 favours predictions that are biased low, we compare models with the Poisson deviance, the natural measure for counts, and also show Neyman and Pearson.

	Neyman		Pearson		deviance		median
model (weights)	255–555	>555	255–555	>555	255–555	>555	dev./N
network ensemble (observed counts)	1.648	1.115	1.554	1.147	1.561	1.121	1.085
free fit (observed counts)	1.581	1.121	1.562	1.223	1.542	1.150	1.096
free fit (Poisson, iterated)	1.643	1.148	1.535	1.103	1.559	1.105	1.075
smooth, $J = 40$ (Poisson)	1.650	1.144	1.542	1.096	1.566	1.100	1.078
smooth, $J = 30$	1.658	1.145	1.549	1.096	1.573	1.101	1.079
smooth, $J = 60$	1.643	1.145	1.535	1.096	1.559	1.100	1.074
smooth, $J = 40$, no threshold factor	1.648	1.144	1.541	1.096	1.564	1.100	1.078

Table 1: Mean per bin on the 1,000 test spectra, below and above 555 GeV, for three measures of agreement with Poisson counts. The weights used in fitting are given in brackets.

- The smooth model is as accurate as the free fit with Poisson weights, and better than the network above 555 GeV (deviance 1.100 vs 1.121, Pearson 1.096 vs 1.147), where the network's biased weights cost it.
- With the observed-count weights the free fit looks better in Neyman's measure (1.581) and worse in Pearson's (1.223 above 555 GeV): the signature of a downward bias.

- The results hardly depend on J (30, 40, 60) and, within 255–1455 GeV, on the threshold factor.

3 Calibration

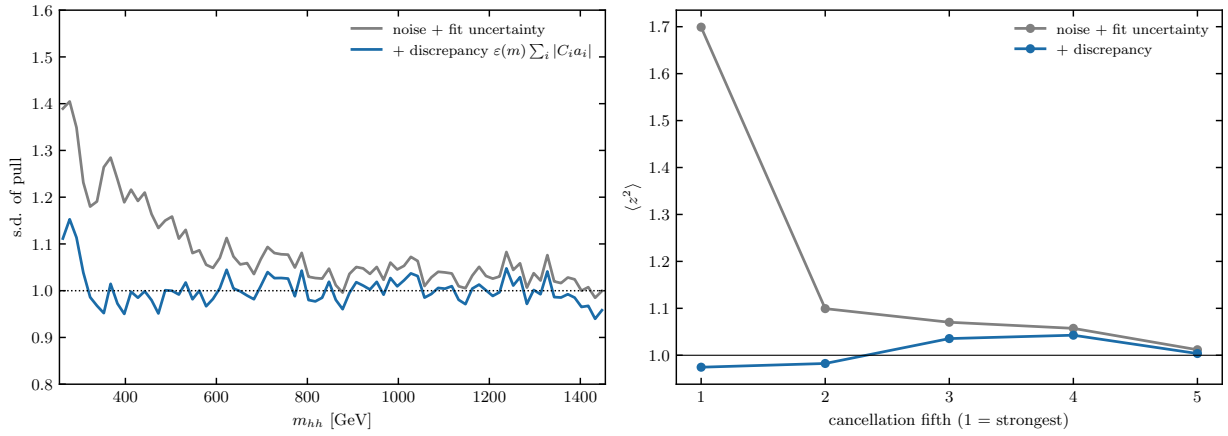


Figure 1: Standard deviation of the pull on the test spectra as a function of m_{hh} (left) and $\langle z^2 \rangle$ by degree of cancellation (right), without and with the discrepancy.

With $z = (t - f)/\sqrt{qf + s^2 + s^2_{\text{disc}}}$, where s^2 is the exact fit variance and $s_{\text{disc}} = \varepsilon(m_{hh}) \sum_i |C_i a_i|$ fitted on the 38,076 quick-test rows: s.d. 1.004 (1.011 below and 1.001 above 555 GeV), 95.4% within ± 2 (95.4% expected), and $\langle z^2 \rangle$ = 0.97, 0.98, 1.04, 1.04, 1.00 from the most to the least cancelling fifth (fig. 1). Without the discrepancy the s.d. is 1.090. The fitted ε rises from 0.08% at threshold to 0.28% at 600 GeV and 3.7% at 1.45 TeV. The fit uncertainty is small: a median 2.9% of the noise, against 5.2% for the free fit (fewer parameters, and smoothness) and 6.1% for the Laplace mixture of the networks (which includes the spread between networks). The total predictive width is a median 1.014 times the noise.

4 Inside a bin: the merged-bin test

The training spectra were merged into 30-GeV bins (pairs of 15-GeV bins), the models were fitted to those, and the 15-GeV bins of the test spectra were predicted. The free fit can only give each half of a pair the same value; the smooth model predicts the split from the shape. The measure is the χ^2 of the difference between the two halves, in units of its noise.

Above about 360 GeV the smooth model fitted on 30-GeV bins predicts the split as well as the model fitted on 15-GeV bins ($\chi^2 \approx 1.0$–1.2), where the halved free fit gives up to 30 (fig. 2). In the first three pairs (255–345 GeV) the split χ^2 is 11, 3.9 and 4.1: the threshold rise has structure on scales below 30 GeV that 30-GeV data cannot fix. Overall, the deviance of the 15-GeV predictions below 555 GeV is 2.34 for the smooth model fitted on 30-GeV bins, 9.35 for the halved free fit and 1.57 for the smooth model fitted on 15-GeV bins.

5 The edges: 240–255 and 1455–1500 GeV

The POWHEG source files contain the bins 240–255 and 1455–1500 GeV, which were not used.

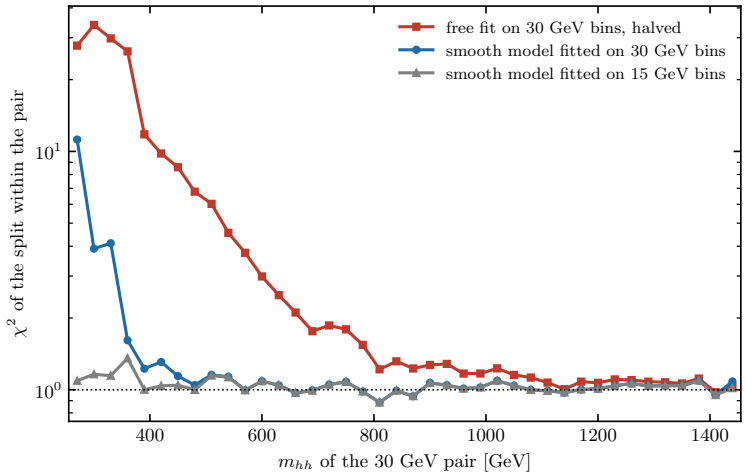


Figure 2: χ^2 of the split of each 30-GeV pair into its two 15-GeV bins on the test spectra. 1 is perfect.

	240–255 GeV	1455–1500 GeV
	observed/predicted, summed over the 1,000 test spectra	
$J = 40$, threshold factor	0.92	1.44
$J = 40$, no threshold factor	0.65	1.42
$J = 30$, threshold factor	0.83	1.05
$J = 60$, threshold factor	0.96	1.07

Table 2: Extrapolation to the unused edge bins.

Below the fitted range, the threshold factor makes the difference: without it the model overpredicts 240–255 GeV by 55%, with it by 4–20%. Above the fitted range the prediction depends on J (5–44% low): there the splines are fixed by the prior alone, and the quoted uncertainties are correspondingly large (pull s.d. 0.3–0.9). **The model should be used within 250–1455 GeV only.**

6 The coefficient functions

Figure 3: the smooth model and the network agree closely for a_1–a_{15} and a_{21}–a_{23}. The poorly determined a_{16}–a_{20} are better determined than in the network (median relative uncertainty 0.74, 0.37, 0.24, 0.21, 0.22 against 0.96, 1.34, 0.23, 0.21, 0.21), and differ from the network where both are uncertain. The roughness of the bin values (rms second difference relative to the function's size) is 0.058, against 0.059 for the network and 0.070 for the free fit. The data determine about 600 combinations of the 920 parameters; for a_1–a_{15} 27–33 of the 38 curvature directions are determined (the smoothness prior matters little), for a_{16}–a_{23} only 12–23 of them.

7 Sensitivity

- **Number of splines**: $J = 30$, 40, 60 give the same accuracy and calibration within 0.01 (table 1). The evidence prefers $J = 40$ (by 47 and 250 units of $\ln Z$ over 30 and 60) in the first pass.
- **Threshold factor**: no effect within 255–1455 GeV; decisive below 255 GeV (table 2). The

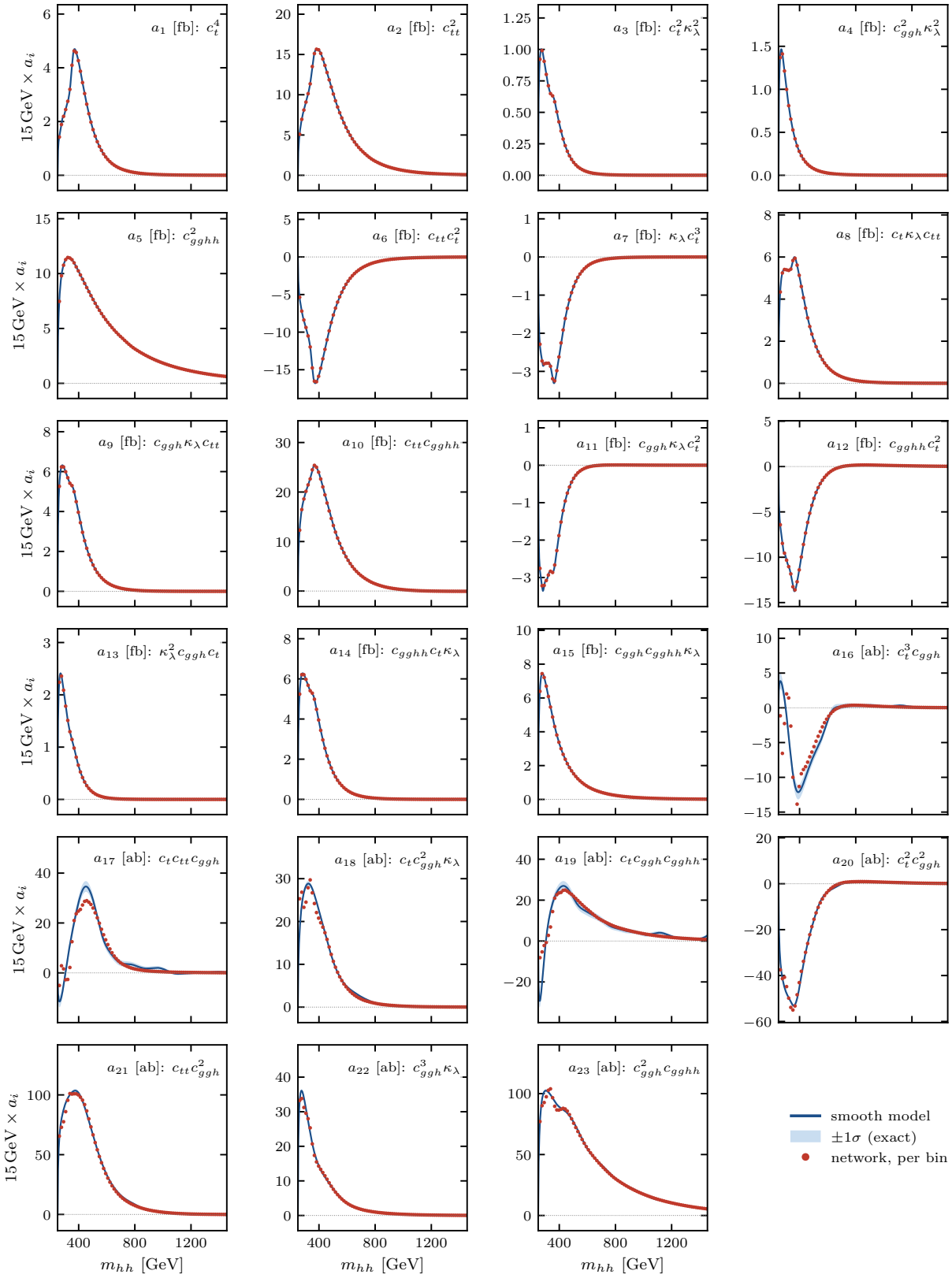


Figure 3: The 23 coefficient functions, as $15\,\mathrm{GeV} \times a_i(m_{hh})$ (the cross section per 15 GeV), with their exact $\pm 1\sigma$ bands, and the per-bin values of the network ensemble (red points).

evidence, computed only on 255–1455 GeV, is slightly higher without it; the edge test is the better guide.

- **Second pass** (the discrepancy in the weights): calibration unchanged (pull s.d. 1.003), but the raw agreement near threshold worsens (deviance 1.87 against 1.57), because the cancelling spectra, which are given less weight, are fitted less closely, and the test spectra share their peculiarity. Since the purpose of the model is to reproduce POWHEG, we recommend the first pass as the primary model, and the difference between the passes as a measure of how much the a_i depend on the treatment of the cancelling spectra.

8 Next steps

1. Make the smooth model the primary model in the paper: rewrite the appendix around it, remake the figures (the 9-panel spectra with the continuous density, the coefficient functions of fig. 3, all test spectra), update the executive summary.
2. Note the weighting bias in the paper, and, for the cross-check, either refit the network fine-tune with Poisson weights or state the bias.
3. The discrepancy correlated in m_{hh} (a Gaussian process), needed for predictions summed over many bins.
4. The questions to the generator side (`docs/modelerror/model_error_note.pdf`), now including the steep rise of ε at high mass.
5. Optionally, finer knots near threshold, which the merged-bin test suggests could help below 345 GeV.

Code. `dihiggs/smooth.py`; `code/smooth_toy.py`; `code/heft_smooth.ipynb` (fit, validation, coefficient functions; writes `runs/SMOOTH_J40/`); the analysis scripts of this note are in `docs/smooth/scripts/`.